

HOUSE PRICE PREDICTION USING MACHINE LEARNING

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Project: House Price Prediction Using Machine Learning

EXECUTIVE SUMMARY

This project developed a machine learning solution for predicting residential house prices using property characteristics such as area, bedrooms, bathrooms, stories, parking, furnishing status, and amenities. Three regression models were evaluated and Gradient Boosting Regressor achieved the best performance with an R^2 score of 67.23%.

DATASET OVERVIEW

- Total Records: 545
- Total Features: 13
- Missing Values: 0
- Duplicate Records: 0
- Target Variable: House Price
- Train-Test Split: 80:20

MODEL PERFORMANCE

Gradient Boosting: MAE 945,253 | RMSE 1,287,074 | R^2 0.6723

Linear Regression: MAE 970,043 | RMSE 1,324,507 | R^2 0.6529

Random Forest: MAE 1,014,947 | RMSE 1,399,769 | R^2 0.6124

TOP 5 PRICE DRIVERS

1. Area (46.2%)
2. Bathrooms (17.3%)
3. Air Conditioning (9.4%)
4. Parking (4.8%)
5. Bedrooms (4.6%)

KEY FINDINGS

- Property area is the strongest predictor of house prices.
- Additional bathrooms significantly increase property value.
- Air conditioning positively influences market pricing.
- Parking availability improves property attractiveness.
- Preferred locations and better amenities contribute to higher valuations.

BUSINESS RECOMMENDATIONS

- Focus on larger residential layouts.
- Include adequate bathroom facilities.
- Improve parking infrastructure.
- Use predictive analytics for pricing decisions.
- Prioritize properties with strong amenities and preferred locations.

CONCLUSION

The Gradient Boosting Regressor delivered the strongest predictive performance and demonstrated the effectiveness of machine learning for real-estate valuation. The project provides actionable insights for buyers, sellers, developers, investors, and real-estate agencies.

FUTURE IMPROVEMENTS

- Geographic location intelligence

- Larger housing datasets
- Hyperparameter optimization
- XGBoost and LightGBM models
- Real-time deployment as a web application